# CHE213
# MASS TRANSFER AND SEPARATION PROCESSES

# GROUP - 12

## EXPERIMENTAL INVESTIGATION OF POSITIVE DEVIATION FROM RAOULT'S LAW


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# AIM

To study vapor-liquid equilibrium (VLE) behavior of a non-ideal binary system of hexane-ethanol and validate modified Raoult's law.

# OBJECTIVE

- Plotting T-x-y and x-y equilibrium diagrams for hexane-ethanol system.
- Calculating activity coefficients ($\gamma_1$, $\gamma_2$) for hexane-ethanol system.
- Analyzing deviation from ideal behavior for hexane-ethanol system.
- Relating molecular interactions to thermodynamic properties.
- Comparing Wilson, NRTL, UNIQUAC models

# MATERIALS AND UTILITIES REQUIRED

**Chemicals:**

- Hexane : 1000 mL (More volatile component).
- Ethanol : 1000 mL (Less volatile component).

**Equipment Setup:**

- Vapor-Liquid Equilibrium (VLE) apparatus (neck flask).
- Heating mantle with temperature control.
- Condenser with cooling water circulation and Water Pump.
- Thermocouples/temperature sensors and Refractometer for analysis.

**Safety Requirements:**

- Face shield, face mask, heat-resistant gloves, and proper ventilation.

# Setup


The setup is the same as the one we used in Vapor-Liquid Equilibrium experiment done in mass transfer lab

The liquids used were

1) Hexane which is non-polar

2) Ethanol which is polar

# EXPERIMENTAL SETUP AND METHODOLOGY

- **Preparation:** Prepared hexane-ethanol mixtures of known compositions (20%, 40%, 60%, 80% hexane).
- **Operation:** Transfered the prepared mixture into the distillation flask and ensured proper sealing to avoid vapor losses.
- Heated the mixture and allowed sufficient time for vapor-liquid equilibrium to establish (steady temperatures).
- **Data Collection:** Recorded bottom temperature (liquid phase) and top temperature (vapor phase).
  Collected condensed vapor and liquid samples.
- **Analysis:** Measured the refractive index (RI) of both samples and converted it to mole fractions using a calibration curve.

# THEORY AND EQUATIONS

- **Raoult's Law** (for ideal solutions): Partial pressure is proportional to mole fraction: $y_i P = x_i P_i^{sat}$
- **Modified Raoult's Law** ( for Non-Ideal solutions): Real mixtures do not follow ideal behavior due to intermolecular interactions. So for Non ideal solutions: $y_i P = x_i \gamma_i P_i^{sat}$ (where $\gamma_i$ corrects for non-ideality).

  $\gamma_i$ = Activity coefficient of component i
- **Deviation:** $\gamma_i > 1$ implies a positive deviation (molecules escape more easily: $P_{real} > P_{ideal}$
- **Antoine Equation:** Used to calculate pure component saturation vapor pressure:

$$\log_{10} P_i^{sat} = A - \frac{B}{C + T}$$

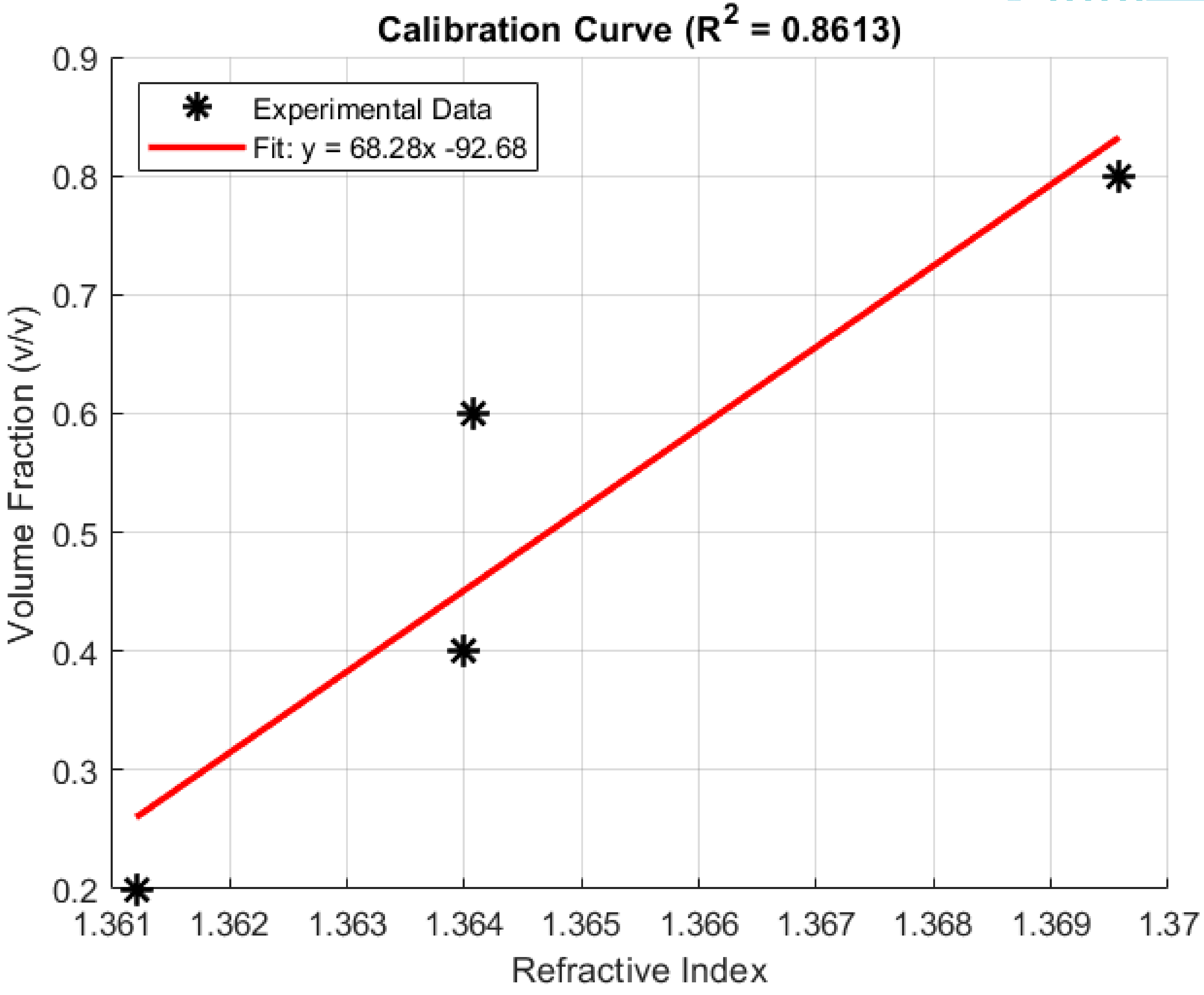
# OBSERVATIONS

| Nominal Comp. | Feed R.I. | Equil. Temp (Vapor / Liquid) | Time(min) | R.I. Vapor (Top) | R.I. Liquid (Bottom) |
|---|---|---|---|---|---|
| **20% Hexane** | 1.3612 | 55.4°C /67.6°C | 0 | **1.36997** | **1.36093** |
| | | 55.7°C /67.8°C | 3 | **1.36949** | **1.36088** |
| | | 55.7°C /67.9°C | 6 | **1.36983** | **1.36048** |
| | | 55.7°C /68.0°C | 9 | **1.36973** | **1.36090** |
| **40% Hexane** | 1.36399 | 55.0°C /59.7°C | 0 | **1.37151** | **1.36382** |
| | | 55.5°C /59.7°C | 3 | **1.37139** | **1.36365** |
| | | 55.7°C /59.7°C | 6 | **1.37142** | **1.36330** |
| | | 55.9°C /59.7°C | 9 | **1.37146** | **1.36341** |
| **60% Hexane** | 1.36408 | 54.7°C /59.1°C | 0 | **1.37087** | **1.36730** |
| | | 54.9°C /59.1°C | 3 | **1.37089** | **1.36714** |
| | | 54.9°C /59.1°C | 6 | **1.37080** | **1.36746** |
| | | 55.0°C /59.1°C | 9 | **1.37096** | **1.36739** |
| **80% Hexane** | 1.36958 | 54.5°C /56.8°C | 0 | **1.37157** | **1.37148** |
| | | 54.7°C /56.8°C | 3 | **1.37162** | **1.37157** |
| | | 54.7°C /56.8°C | 6 | **1.37157** | **1.37148** |
| | | 54.7°C /56.9°C | 9 | **1.37171** | **1.37152** |

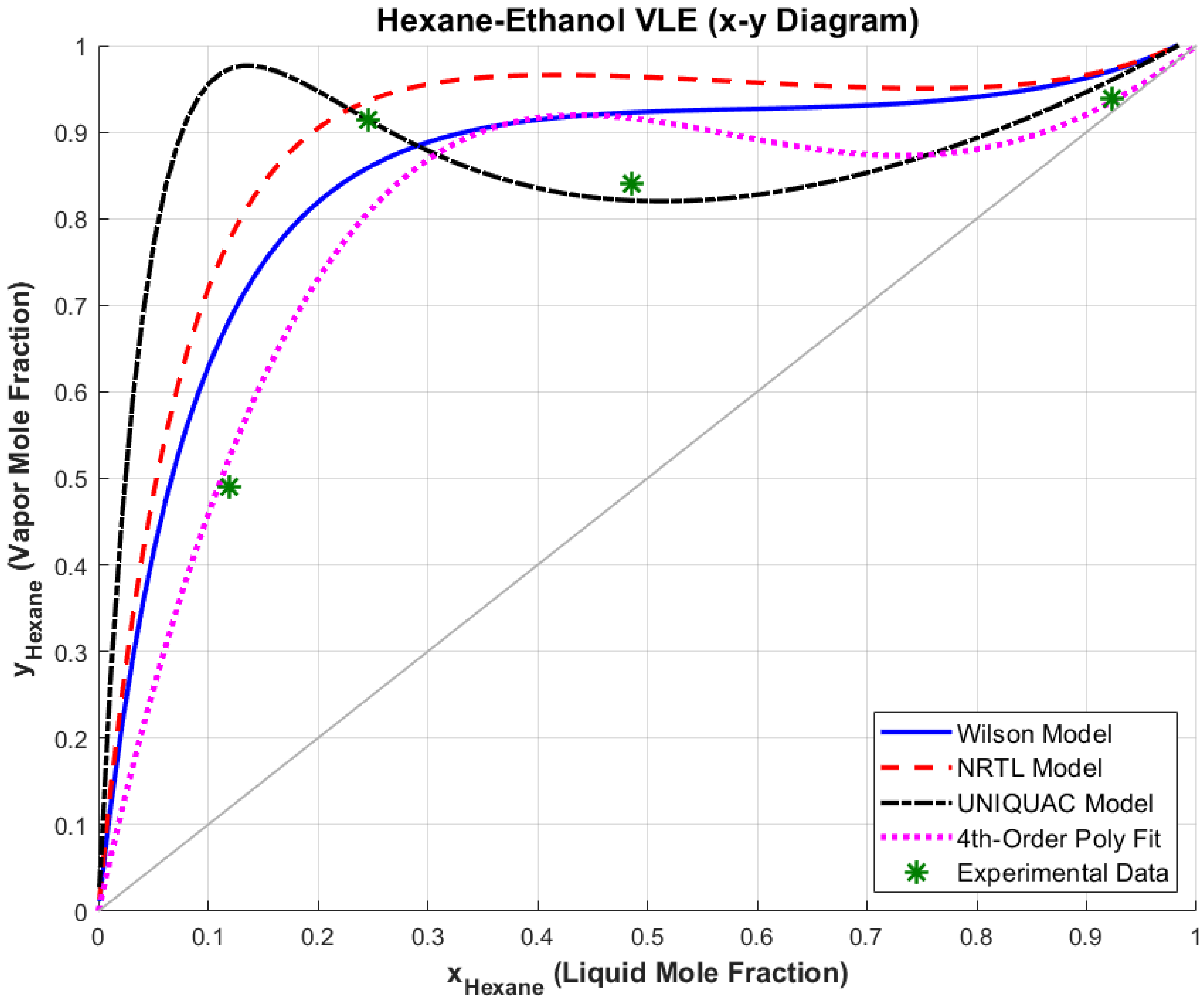
# CALIBRATION CURVE

**Calibration Curve:** Generated using standard known mixtures to relate Refractive Index (RI) to Volume Fraction.

- Linear Fit Equation: y = 68.28x - 92.68 (with an R^2 value of 0.86).



# EQUILIBRIUM CURVE




**x-y Equilibrium Diagram:**
Plots the Liquid Mole Fraction (x) vs. Vapor Mole Fraction (y).

- Observation: The green experimental data points sit above the ideal y=x line.

This confirms that the vapor phase is always richer in the more volatile hexane component.

# WILSON MODEL

The Wilson model is particularly effective for describing the non-ideal behavior of polar/non-polar mixtures like Hexane and Ethanol by accounting for the difference in intermolecular forces and molecular size.

Model Parameters used in Analysis:

- Molar Volumes (V_i):
  - V_Hexane = 131.61 cm³/mol
  - V_Ethanol = 58.68 cm³/mol
- Binary Interaction Parameters (Lambda_ij):
  - a_12 = 437.98 cal/mol,
  - a_21 = 1438.0 cal/mol,

**The Governing Equations:**

$$\ln \gamma_1 = -\ln(x_1 + \Lambda_{12}x_2) + x_2 \left( \frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{\Lambda_{21}x_1 + x_2} \right)$$

$$\ln \gamma_2 = -\ln(x_2 + \Lambda_{21}x_1) - x_1 \left( \frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{\Lambda_{21}x_1 + x_2} \right)$$

$$\Lambda_{ij} = \frac{V_j}{V_i} \exp\left( -\frac{a_{ij}}{RT} \right)$$

# NRTL MODEL

TNRTL is a versatile model applicable to both VLE (Vapor-Liquid) and LLE (Liquid-Liquid) equilibria. It introduces a "non-randomness" factor to account for how molecules prefer to cluster.

Model Parameters used in Analysis:

- Non-randomness Constant (alpha): 0.47
- Energy Interaction Parameters (Delta g_ij):
  - g_12 = 1036.0 cal/mol
  - g_21 = 1095.0 cal/mol

**Governing Equations:**

$$\ln \gamma_1 = x_2^2 \left[ \tau_{21} \left( \frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{\tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2} \right]$$

$$\ln \gamma_2 = x_1^2 \left[ \tau_{12} \left( \frac{G_{12}}{x_2 + x_1 G_{12}} \right)^2 + \frac{\tau_{21} G_{21}}{(x_1 + x_2 G_{21})^2} \right]$$

$$\tau_{ij} = \frac{\Delta g_{ij}}{RT}$$

$$G_{ij} = \exp(-\alpha \tau_{ij})$$

# UNIQUAC MODEL

UNIQUAC splits the activity coefficient into two parts: a combinatorial part (based on molecular shape/size) and a residual part (based on energy interactions).

Model Parameters used in Analysis:

- Structural Parameters (r, q):
- Hexane: r = 4.4998,
  - q_1 = 3.856
- Ethanol: r_2 = 2.1055,
  - q_2 = 1.972
- Interaction Energies (u_ij):
  - a_12 = 328.34 cal/mol
  - a_21 = 339.74 cal/mol
- Coordination Number (z): 10

## Fundamental Equations:

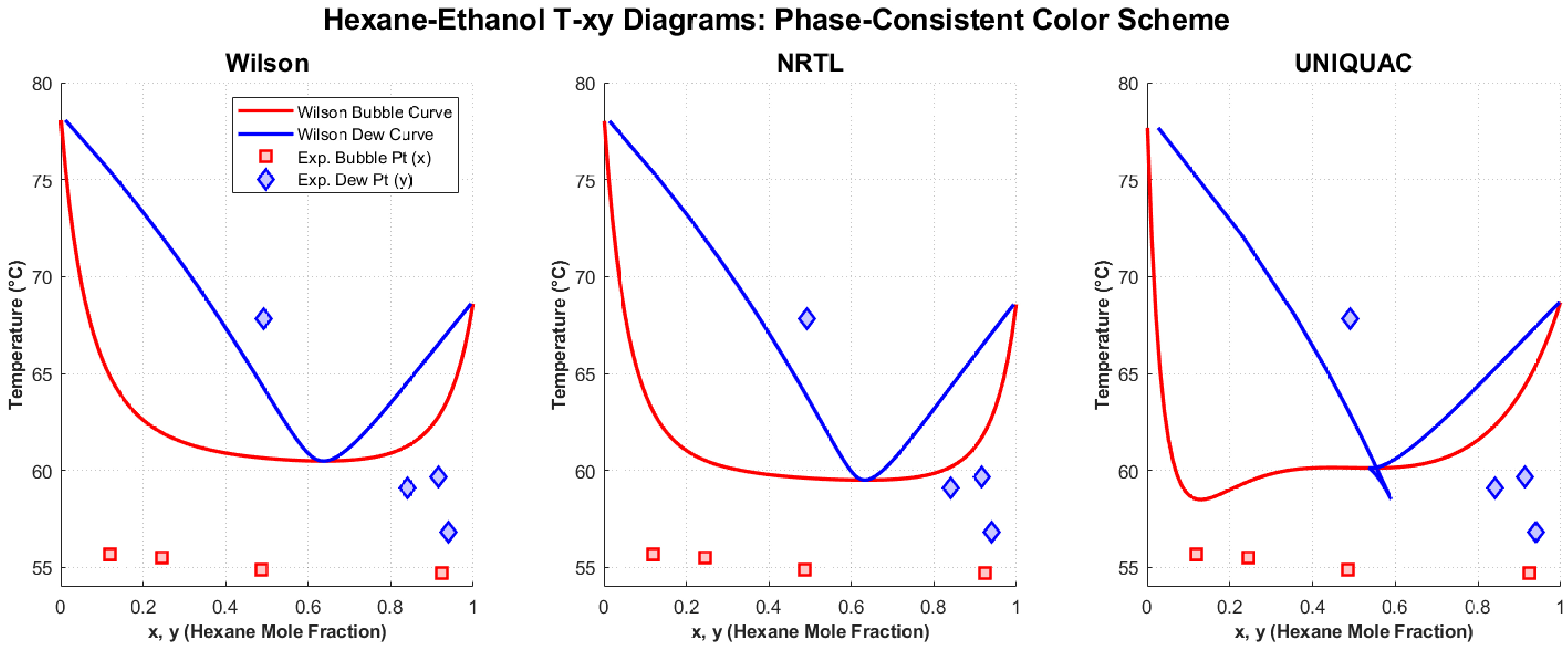
$$\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R$$

$$\ln \gamma_i^C = \ln \frac{\Phi_i}{x_i} + \frac{z}{2} q_i \ln \frac{\theta_i}{\Phi_i} + L_i - \frac{\Phi_i}{x_i} \sum x_j L_j$$

$$\ln \gamma_i^R = q_i \left[ 1 - \ln \left( \sum \theta_j \tau_{ji} \right) - \sum \frac{\theta_j \tau_{ij}}{\sum \theta_k \tau_{kj}} \right]$$

# T-xy CURVE

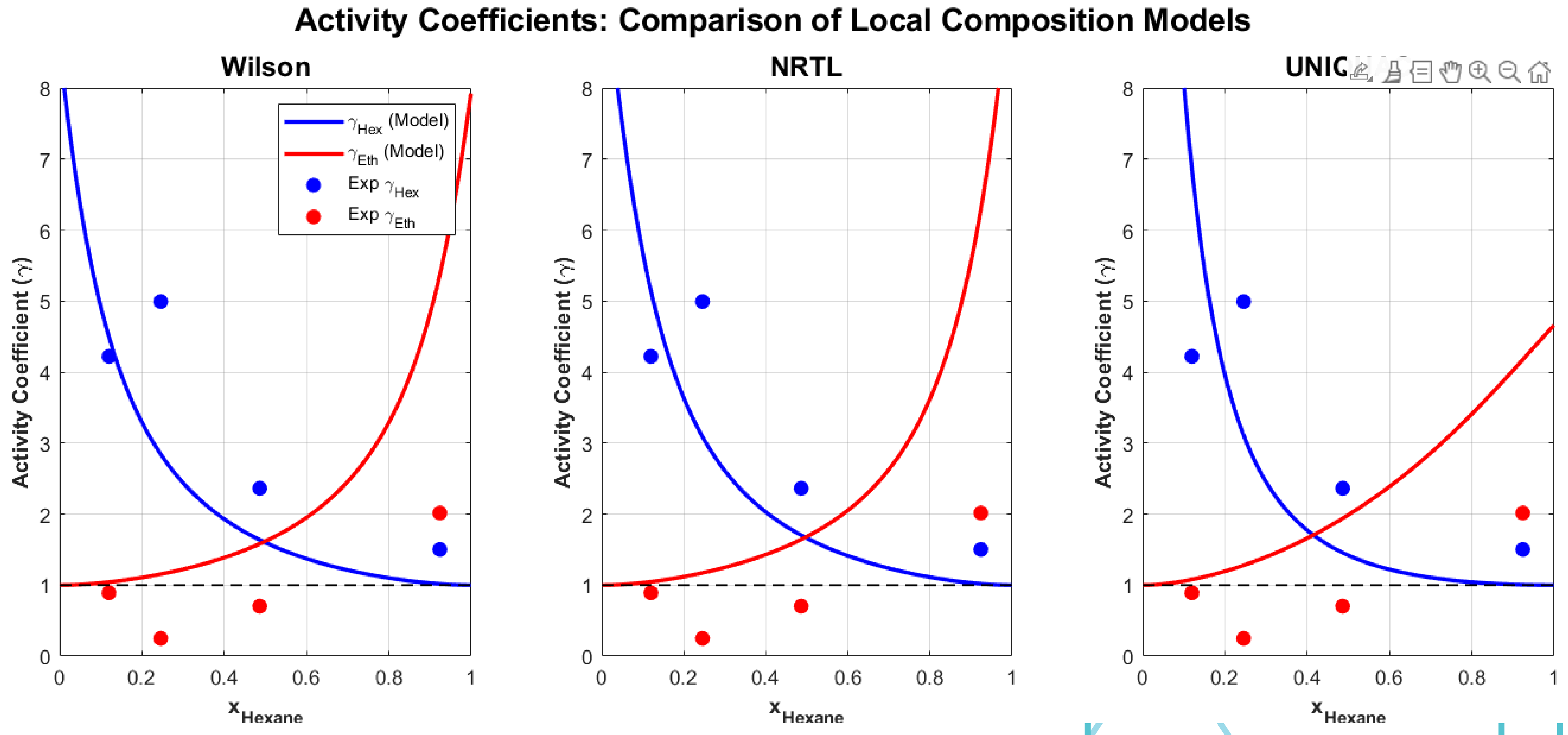
- **Bubble and Dew Points:** The solid lines represent the Bubble Point (liquid starting to boil). And the dashed lines represent the Dew Point (vapor starting to condense).
- **Model Comparison:** The three thermodynamic models (Wilson, NRTL, and UNIQUAC) successfully map the wide phase envelope, visually demonstrating how temperature changes with varying liquid (x) and vapor (y) compositions.



# ACTIVITY COEFFICIENTS

- **Quantifying Non-Ideality:** The graphs map the calculated activity coefficients ($\gamma$1 for hexane, $\gamma$2 for ethanol) across the mole fraction spectrum.
- **Proof of Positive Deviation:** For all compositions, the activity coefficients remain above 1 ($\gamma$ > 1).

This is due to the weak interactions between the highly polar ethanol molecules and the non-polar hexane molecules, allowing them to escape the liquid phase easier than predicted by ideal models.



# CONCLUSION

- **Successful Validation:** The experiment successfully captured the VLE data for the non-ideal Hexane-Ethanol system.
- **Positive Deviation Confirmed:** Both the experimental data and the theoretical modeling confirmed a strong positive deviation from Raoult's Law ($\gamma > 1$).
- **Thermodynamic Modeling:** The Wilson, NRTL, and UNIQUAC models were successfully applied to the experimental data, accurately mapping the T-x-y envelope and x-y equilibrium curves.
- **Phase Enrichment:** As expected, the lower boiling point component (Hexane, 69°C) successfully enriched the vapor phase across all mixture concentrations.

# Thank You

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